



Thermal interface materials with low modulus and high thermal conductivity by manipulating multi-branched PDMS network: The disentanglement and entanglement effect of dangling chain

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ABSTRACT

Thermal interface materials, which are used to fill between a heat sink and a heat-generating device to reduce the thermal resistance, are required to integrate low modulus and high thermal conductivity. However, the coexistence of these seemingly contradictory properties between modulus and thermal conductivity remains a persistent challenge. Here, this work reported thermal interface materials with low modulus and high thermal conductivity by manipulating multi-branched polydimethylsiloxane (PDMS) network. The competition between entanglement and disentanglement of dangling chain has been fully studied on pure PDMS networks and the composites. When the PDMS dangling chain spacing is set at 149 atoms, the modulus reaches a remarkably low value of 21.2 kPa by disentanglement. Furthermore, this thermal interface materials filled with 92 wt% aluminum, exhibits low modulus (0.39 MPa), high elongation at break of 94 % and outstanding thermal conductivity (6.02 W/(m K)). The synergistic combination of exceptional thermal conductivity and superior mechanical properties in these elastomers opens up new avenues for their applications in thermal management.

1. Introduction

Thermal interface materials (TIMs) are used to fill between a heat sink and a heat-generating device to reduce thermal resistance [1,2]. In order to ensure the reliability of the normal operation of the chip, the thermal interface material needs a high thermal conductivity to quickly transfer heat to the heat sink and radiator [3–6]. Thermal interface materials are made of different materials for different performance requirements of chip work. At present, the pursuit of high thermal conductivity and strength and toughness in thermal interface materials [7–11]. In addition, during the heating and cooling of chip, the thermal stress and the deformation generated due to the different expansion coefficients of materials [12–15]. Low modulus and high ductility of TIMs are anticipated properties to protect the chip [16–18]. The surface of the chip and the heat sink are not smooth, the rigid contact leaves thermally insulating air gaps. Thus, low modulus TIM is better to fill the interface and reduce the thermal interfacial resistance. However, the coexistence of these seemingly contradictory properties between modulus and thermal conductivity remains a persistent challenge for

existing TIMs [19–22]. In order to have a high thermal conductivity, thermal interface materials need to be filled with more fillers, but the increase of fillers will lead to an increase in the Young's modulus of the material. Previous reports have emphasized the relationship between modulus and thermal conductivity. Li et al demonstrated the addition of filler can make the thermal conductivity increase exponentially, the more filler is added, the density of filler per unit volume increases, and the thermal conductivity increases exponentially. However, when the thermal conductivity increases, its modulus also increases by a factor of 20–30 [23,24]. For composites, the module depends on the filler's ratio and polymer's module. In order to ensure the high thermal conductivity, polymer plays a key role to reach low module TIMs [25–27]. For polymer matrices, brush polymers have extremely low modulus compared to conventional polymers. The dangling chains of the brush-like polymer change the conformation [28–30], increase the steric hindrance of the polymer matrix and untangle the tangle between the backbones, and the polymer matrix also exhibits low modulus and soft properties. Singh et al researched on the controlled growth of numerous branched chains on the polymer backbone has led to the development of a series of ultra-

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flexible brush polymer network structures, featuring a modulus as low as 10^4 Pa [31]. Ding et al developed a comb-brushed PDMS network with short dangling ends. And the relevant composite exhibited high thermal conductivity (4.9 W/(m K)) and excellent damping capacity [32]. Meanwhile, He et al constructed brush-shaped polymer and the composite with ultralow elastic modulus. Their result proved dangling chain is benefit to decrease modulus [33]. However, there have been no reports to study the effect of side chain spacing on polymers and TIMs with constant crosslinking density.

It is noteworthy that, at a certain extent, the presence of dangling chains in multi-branched polymers leads to interactions between dangling chain and dangling chain, dangling chain and filler (Fig. 1b). The formed physic entanglement will impede the movement of chain segments, resulting in an increase in the rigidity of the polymer backbone and a subsequent rise in modulus. To study the competition between disentanglement and entanglement effect of dangling chain, this work chooses (vinylmethylsiloxane)-dimethylsiloxane copolymer (multi-vinyl-PDMS) as backbone, dihydride-terminated poly(dimethylsiloxane) (DH-PDMS) as covalent crosslinks and monohydride-terminated poly(dimethylsiloxane) (MH-PDMS) as dangling chain to construct multi-branched network (Fig. 1a). The suitable dangling chain spacing shows disentanglement effect, making PDMS₁₄₉ has the lowest modulus (21.2 kPa). Furthermore, aluminum powders (50.0 μm , 15.0 μm and 2.0 μm) were chosen to establish an effective thermal conductivity pathway within the composite material (Fig. 1b). The mass fraction of fillers was kept at 92 wt% unless otherwise specified. Our result exhibits that the dangling chain density is a key factor in multi-branched network to prepare thermal interface materials with low modulus (0.39 MPa), high elongation at break of 94 % and outstanding thermal conductivity (6.02 W/(m K)).

2. Results and discussions

2.1. Design strategy of PDMS and thermal interface materials

To precisely control the dangling chain density, multi-vinyl-PDMS has the same molecular weight ($M_w = 28$ K) but different vinyl contents. Nuclear magnetic resonance (NMR) analysis was used to calculate the vinyl content and the spacing X (X refers the atoms between two vinyl group). The result was illustrated in Fig. S1 and detailed in Table S1. The regulation of the cross-linking network was achieved by maintaining a constant mass ratio between the backbone and the cross-linked chain. The determination of side-chain spacing involved

calculations based on the molecular weight of the backbone and the quantitative molar mass obtained through NMR analysis. Here, the side-chain spacing was defined as the distance between each pair of vinyl groups on the main chain in terms of the number of atoms. This approach aimed to fulfill the experimental design criteria of maintaining an unaltered cross-linking density while varying the side-chain spacing. In terms of filler network design, aluminum powders of different particle sizes are added to form a good thermal conductivity pathway, as depicted in Fig. 1b.

2.2. Mechanical properties and theoretical analysis of multi-branched PDMS

To examine the mechanical properties of the polymer networks with varying dangling chain spacing, this work carried out stress-tensile experiments (Fig. 2a). The results indicated that the polymer network, augmented with additional dangling chains, exhibited commendable Young's modulus and elongation at break, rendering it suitable as a matrix for the fabrication of thermal interface materials. Fig. 2b illustrates that the Young's modulus of the polymer networks exhibited a declining-then-increasing trend in response to a reduction in dangling chain spacing. Notably, at a dangling chain spacing of 149, the network exhibited the lowest modulus (21.2 kPa) and the highest elongation at break (549 %). To delve into the underlying mechanism, this work conducted frequency scans on networks with different dangling chain spacing (Fig. 2c). The storage modulus initially decreased and then increased with decreasing dangling chain spacing. The disentanglement of dangling chain was observed at PDMS₁₄₉, followed by entanglement effect at shorter dangling chain spacing. In Fig. 2d, The disentanglement and entanglement effect of dangling chain can be further verified by test its complex viscosity at 0.05 rad s^{-1} ($\eta_{0.05}^*$) [34]. Compared to PDMS_{cross-linked}, PDMS₁₄₉ has the lower value indicating disentanglement of dangling chain. PDMS₄₆ and PDMS₂₆ show entanglement effect. Furthermore, the molecular weight (M_e) of the tangled points according to storage modulus was employed to evaluate the entanglement of PDMS networks (Formula S1, Supporting Information). The value was calculated and displayed in Fig. 2e. The result is agreed well with the conclusion above. That is the suitable dangling chain content is conducive to disentanglement, while entanglement occurs and increases with more dangling chain.

To further elucidate the network structure, we derived the relationship between the relaxation spectrum and relaxation time by incorporating the storage modulus and loss modulus from Fig. 2c into

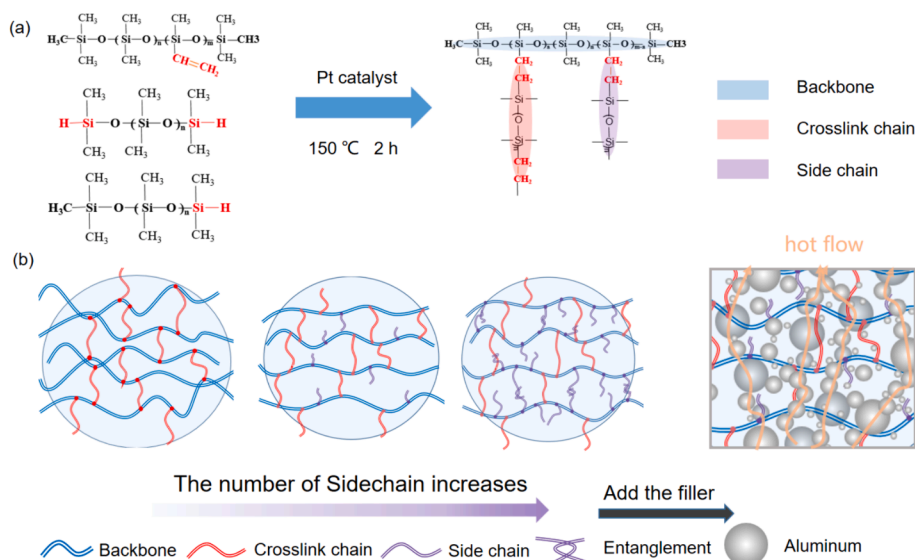


Fig. 1. Design of PDMS and thermal interface materials. (a) preparation of PDMS, (b) elastomers vary with the sidechain spacing and TIMs.

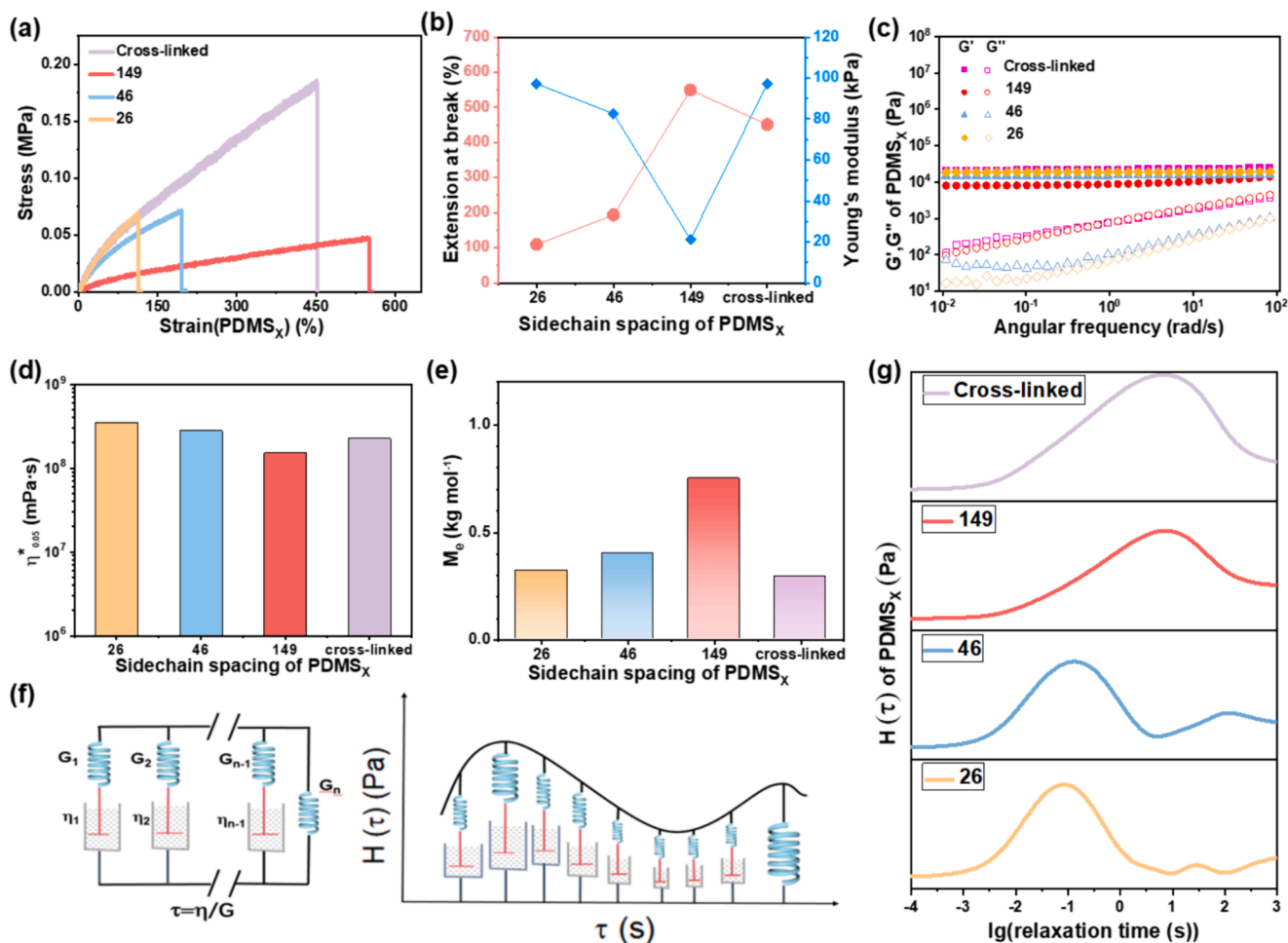


Fig. 2. Mechanical properties and theoretical analysis of multi-branched PDMS_x (X is sidechain spacing, X = 26, 46, 149, cross-linked) elastomers. (a) Stress–strain curves and. (b) Young's modulus and elongation at break and (c) frequency-dependent storage modulus (G') and loss modulus (G'') and. (d) Complex viscosity and. (e) molecular weight between entanglement points with different sidechain spacing. (f) Each Maxwell element consists of a spring and a punch connected in series; the relaxation spectrum $H(\tau)$ can be described by a set of spring stiffnesses and relaxation times. (g) Relaxation spectra ($H(\tau)$) for the PDMS with different sidechain spacing.

Maxwell's model (Fig. 2f) and the following formula [35]:

$$G' = \int_{-\infty}^{+\infty} H(\tau) \frac{\omega^2 \tau^2}{1 + \omega^2 \tau^2} d\ln(\tau) \quad (1)$$

$$G'' = \int_{-\infty}^{+\infty} H(\tau) \frac{\omega \tau}{1 + \omega^2 \tau^2} d\ln(\tau) \quad (2)$$

where the $H(\tau)$ value is equal to the energy expended for dynamic shock loading and unloading at a given τ measurement frequency, which can also be understood as the dynamic energy generated by the series of Maxwell's elements in series in Fig. 2f.

The energy dissipation and relaxation time of PDMS with the change of side chain distance can be seen in Fig. 2g. Combined with Fig. 2e, the molecular weight of the cross-linked chains first increases and then decreases with the decrease of side chain distance, so the relaxation time first increases and then decreases.

2.3. Mechanical properties of elastomer composites-TIMs

The different particle size of aluminum powder has influence on its mechanics and the thermal performance. The introduction of large particle size can form a good thermal conductivity path and greatly improve the thermal conductivity, but the toughness of the material also decreases. The introduction of small particle size can increase the

toughness of the material, and the introduction of different particle sizes can make better use of the filling space and form more thermal conductivity channels. Aluminum powders (50.0 μm , 15.0 μm and 2.0 μm) were chosen to balance the thermal and mechanical properties. In our system, disentanglement and entanglement effect of dangling chain have been observed in polymer networks by varying the dangling chain content. The enhancement of thermal conductivity is primarily achieved through the incorporation of fillers. The interaction between polymer chain and fillers was generated. To address this interplay, we examined the tensile curve and thermal conductivity of TIM₁₄₉ with varying mass fractions (refer to Fig. S2 and Fig. S3 in the Supporting Information). In the following experiment, 92 wt% of filler was choose to scrutinize the properties and underlying mechanisms of TIMs with different dangling chain content. Fig. 3a illustrates the viscosity tests of uncured TIMs. The addition of dangling chains results in a decreased viscosity, and the entangled region of the polymer chain falls within the range of 0.01–0.1 s^{-1} . The observable trend indicates a more pronounced platform with higher dangling chain additions, signifying the presence of more entanglements before the curing process[29]. Simultaneously, Fig. 3b presents the viscosity values at 5 s^{-1} for different dangling chain spacing, all of which are notably low (<400 Pa s). Lower viscosity values imply a greater capacity for filler incorporation and dispensing application. In Fig. 3c, the stress–strain curves of TIMs with varying dangling chain spacing are depicted. Upon the addition of 92 wt% filler, the

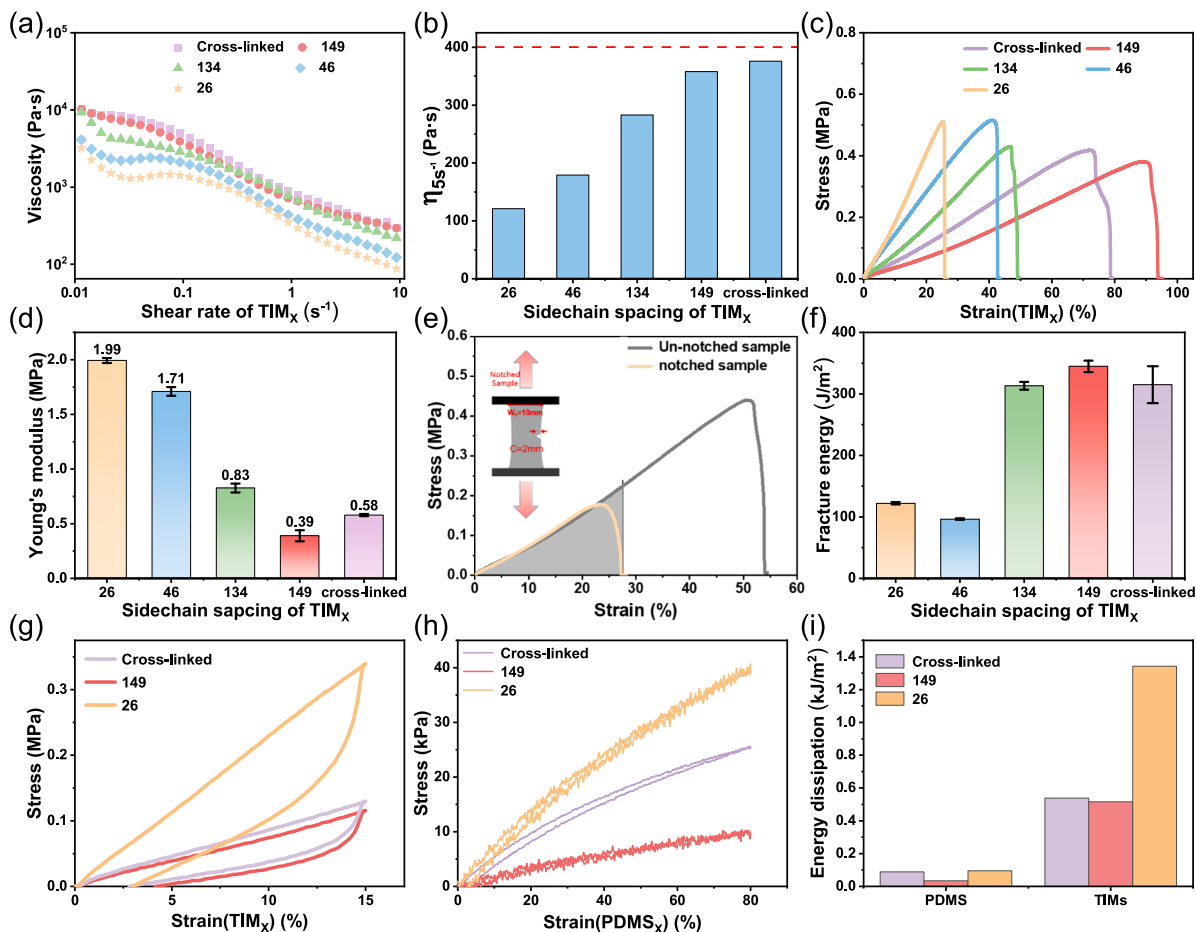


Fig. 3. Mechanical properties of elastomer composites-TIM_x (X is sidechain spacing, X = 26, 46, ...). (a) The viscosity varies with the shear rate and (b) viscosity at a shear rate of 5 s⁻¹ and (c) stress-strain curves and (d) Young's modulus and. (e) Notched and un-notched sample tensile stress-strain curves and (f) the fracture energy of the TIMs with different sidechain spacing. (g) mechanical stress dissipation of TIMs. (h) mechanical stress dissipation of PDMS. (i) energy dissipation of PDMS and TIMs.

modulus of TIM experiences a substantial increase, approximately 20 times higher than that of polymer network. Because of the interfacial interaction between dangling chain and fillers forming physic cross-linking, the stress and modulus increase obviously. The values of TIMs with dangling chain are higher than TIM_{cross-linked}, except TIM₁₄₉. Notably, the competition between the disentanglement of dangling chain and the increase of physic crosslinking makes TIM₁₄₉ has the smallest modulus at 0.39 MPa. In addition, Fig. 3e is a single edge notched experiment to test the toughness of the TIM (Fig. S4, Supporting Information), calculated from the stress-strain curves of the notched sample and the notched sample, and the calculation formula [35] is:

$$\Gamma_c = \frac{6wc}{\sqrt{1+\lambda}} \quad (3)$$

where λ is the elongation of the notched specimen during crack propagation, w is the integration interval of the gray area in Fig. 3d, and c is the notch width. As shown in Fig. 3f, the maximum toughness of TIM₁₄₉ is 344.8 J/m². The toughness decreased obviously with the dangling chain content increase, reaching a minimum toughness of 96.4 J/m² for TIM₄₆.

Concurrently, an analysis of the tensile curves of PDMS and TIMs reveals noticeable variations in modulus, especially in the absence of cross-linking and with a side-chain spacing of 149. Subsequently, the energy dissipation of PDMS and TIMs is individually calculated. Tensile

cycles for PDMS and TIMs are depicted in Fig. 3g and h, respectively, where the enclosed area of the cyclic curves signifies their energy dissipation capacity. Fig. 3i provides a comparative overview of the energy dissipation of PDMS and TIMs. The energy dissipation of TIMs, particularly with dangling chains, increases following the addition of filler. The collaborative effect of dangling chains and fillers is more pronounced.

2.4. Theoretical analysis of modulus and energy dissipation

To disclose the influence of polymer structures with different dangling chain spacing on the mechanical properties of TIM, the stress-strain curves can be fitted with viscoelastic models [34].

$$\sigma(\lambda) = \left(\frac{2G_v D_e}{1-2D_e} \left(1 - \exp\left(\frac{-1-2D_e}{D_e} (\lambda-1) \right) \right) \right) + \frac{G_v D_e}{1-2D_e} \left(1 - \exp\left(\frac{-1+D_e}{D_e} (\lambda-1) \right) \right) \right) \lambda^{-1} + \left(\frac{G_e}{1 - \frac{\lambda^2 + 2\lambda^{-1} - 3}{\lambda_m^2 + 2\lambda_m^{-1} - 3}} (\lambda^2 - \lambda^{-1}) \right) \lambda^{-1} \quad (4)$$

where λ_m represents the theoretical finite extensibility of polymer chains, G_v is the initial shear modulus of the viscoelastic part, G_e is the shear modulus of the elastic part at a small strain, and D_e is the Deborah

number. The stretch ratio λ is related to the tensile strain ε as $\lambda = \varepsilon + 1$. The sum of G_v and G_e is equal to the shear modulus of the gel at high frequencies. Fig. 4a illustrates the stress–strain curves and fitting curves for TIM₂₆ and TIM_{cross-linked}, with the fitting curves of the model aligning well with the actual tensile curves. The parameters derived from the fitting curve enable the estimation of α (Formula S2-S4, Supporting Information), the Kuhn chain length (b_k), the Kuhn number (N_p) for each part of the chain, and the end-to-end distance of the chain segment. In Fig. 4b, it is observed that the excessive addition of filler hinders the movement of the chain segment, resulting in a larger b_k , a smaller N_p , and relatively stable b_k values with varying dangling chain spacing. Notably, N_p attains its maximum value when the dangling chain spacing is 149. The mean square radius of rotation of TIMs can be computed using b_k and N_p (Fig. 4c), where TIM₁₄₉ exhibits the largest mean square radius. This implies a higher degree of freedom for the polymer chain segment and superior flexibility, consequently leading to the lowest modulus. Furthermore, Fig. 4d serves as a tool to assess the respective contributions of entanglement and chemical cross-linking within the polymer network to the mechanical properties of the TIMs. We use the middle part of the stretch curve to fit into the Rubinstein–Panyukov formula [27,31,35],

$$\frac{\sigma_{eng}}{\lambda - \lambda^{-2}} = G_x + \frac{G_e}{0.74\lambda + 0.61\lambda^{-1/2} - 0.35} \quad (5)$$

The midsection of the stretching curve corresponds to the polymer disentangling process. Our experimental observations confirm that the chemical crosslinking (G_x) in TIMs aligns with expectations, and the density of the crosslinking network remains constant. In the absence of dangling chains, the physical entanglement (G_e) measures 80.4 kPa. The introduction of dangling chains, specifically with a spacing of 149, facilitates untangling, leading to an increase in entanglement as dangling chain spacing decreases. Fig. 4e displays the storage modulus and loss

modulus of TIMs, showcasing constancy despite variations in frequency. In comparison to PDMS, TIMs exhibit a distinct plateau modulus, with filler additions enhancing polymer chain interactions, the polymer chain will be adsorbed on the small particle size aluminum powder, the relaxation time of the filler is longer, and the stress relaxation phenomenon of the composite material is more obvious, so there is a platform modulus that the matrix does not have (Fig. S5 and Fig. S6, Supporting Information). The SEM analysis of TIM₁₄₉ (Supporting Information Fig. S5) indicated that the aluminum hybrids are uniformly dispersed and stability. Fig. 4f illustrates the relaxation spectrum of TIMs, revealing three Gaussian peaks in each spectrum. From right to left, peaks between chemically cross-linked chains endure from 10 s to 10² s with no alteration. The interaction between dangling chains, around 1 s, witnesses increased energy dissipation as dangling chain spacing decreases, elucidating the rise in material entanglement with reduced dangling chain spacing and, consequently, increased modulus. The leftmost peak represents the interaction between the chain segment and the resin, slightly attenuated at TIM₁₄₉ due to dangling chain introduction, which widens the spacing between main chains, diminishing the interaction with the resin to a certain extent. To gain further insights into chain segment dynamics, Fig. 4g and h present dipole reorientation data from a wide-band dielectric spectrum test conducted at −100 °C, facilitating a comprehensive study of polymer dynamics. When normalized within the 10–10² s range, in this dynamic range, dipole fluctuations will occur in the side chain, and charge transfer and polarization effects will occur at the interface of the inner and outer layers, which determine the dielectric properties of the material, thus reflecting the characteristic structure information of PDMS. Fig. 4h illustrates the original curves of TIM₂₆ and TIM_{cross-linked}. In the case of TIM₂₆, segment relaxation predominantly involves the movement of high-frequency segments. Conversely, the relaxation of TIM_{cross-linked} segments is characterized by a greater extent of chain movement

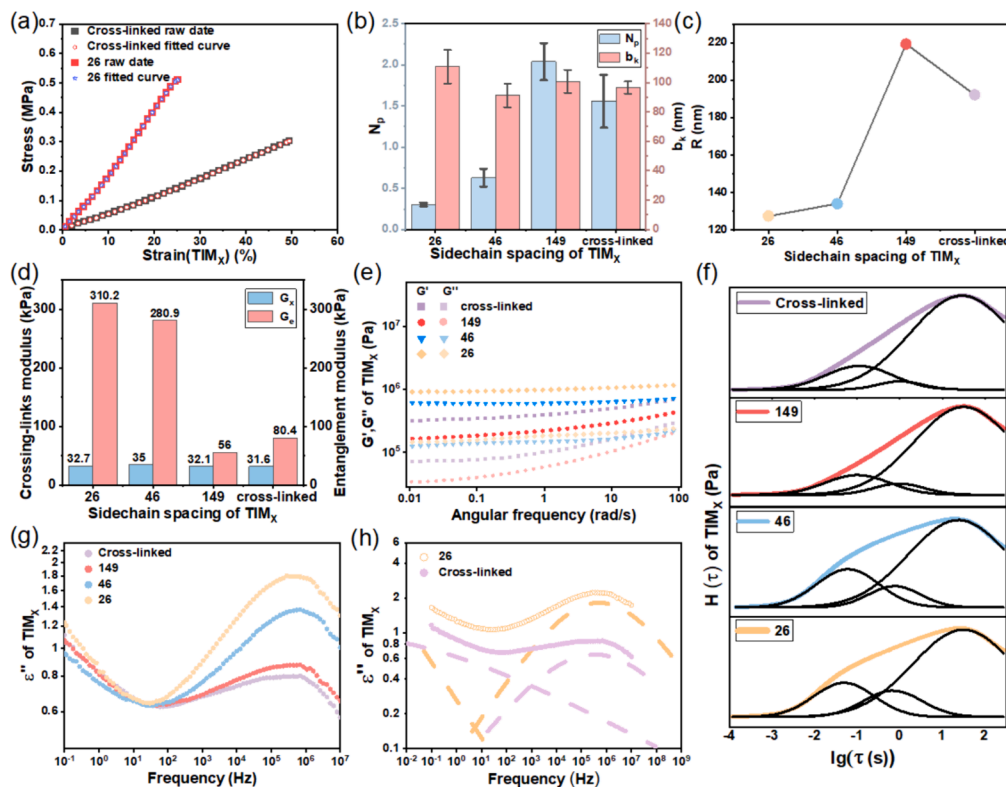


Fig. 4. Theoretical analysis of modulus and energy dissipation. (a) fitted curve and raw date of TIM₂₆ and TIM_{cross-linked}. (b) Kuhn number of partial chain (N_p) and the Kuhn length (b_k) and (c) mean square radius of gyration (d) Physical entanglement modulus and chemical cross-links modulus of the TIM_x (X is sidechain spacing, X = 26,46,...) with different dangling chain spacing. (e) frequency-dependent storage modulus (G') and loss modulus(G'') and (f) Relaxation spectra ($H(\tau)$) and (g) broadband dielectric spectrum (−100 °C) for the TIMs with different sidechain spacing. (h) Dielectric loss for TIM₂₆ and TIM_{cross-linked}.

occurring at low frequencies.

2.5. Thermal conductive properties and application of thermal interface materials

The primary function of the TIMs is the expeditious transfer of heat from the hot terminus to the cold terminus. This process involves traversing three distinct stages, encompassing the thermal resistance inherent at both the hot and cold termini, the TIMs themselves, and the thermal resistance intrinsic to the TIMs. It is imperative to minimize the thermal resistance across both interfaces, necessitating the formulation of TIMs possessing a low modulus to impeccably conform to the interface, thereby mitigating the presence of air, diminishing air heat transfer, and augmenting overall heat transfer efficiency.

The thermal conductivity and interfacial thermal resistance were obtained by ASTM D5470 standard test. In detail, we measured the three total thermal resistance with three different thicknesses, which are used to fit the curve. The reciprocal of the slope is the thermal conductivity, and the intercept of the ordinate is the interfacial thermal resistance (taking TIM₁₄₉ as a sample, Supporting Information S7). Fig. 5a delineates the thermal conductivity and interfacial thermal resistance characteristics of TIMs exhibiting diverse dangling chain spacing. Notably, in instances of TIM₁₄₉ with a considerable interchain distance, the aluminum powder can be more effectively laden and uniformly dispersed. TIM₁₄₉ manifests a notable thermal conductivity of 6.02 W/(m K) and a low thermal resistance of 0.104 (cm² °C)/W under a pressure of 40 psi. Moreover, the interfacial thermal resistance has the same trend as the modulus of TIM. The minimum interfacial thermal resistance of TIM₁₄₉ was 0.555 (cm² °C)/W because of the lowest modulus

(Fig. S8, Supporting Information). In addition, the thermal interface material demonstrates commendable stability at an elevated temperature of 150 °C (Fig. S9, Supporting Information). To underscore that the efficacy of TIMs warrants practical validation in real-world applications, Fig. 5b illustrates the heat dissipation performance of blank, TIM₂₆, and TIM₁₄₉ when subjected to a 10 W power LED lamp. As anticipated, TIM₁₄₉ exhibits superior heat dissipation capabilities, maintaining a temperature of only 41.1 °C after 60 s of power-on—well within the safe operating temperature range for the LED lamp. Given the current predominant application of TIMs in chip heat dissipation, we conducted simulations emulating the chip operating environment, as depicted in Fig. 5c, to delve deeper into their heat dissipation efficacy. In Fig. 5d, a power density of 27–30 W/(cm²) is applied for 1200 s to evaluate the performance of Without TIM, Commercial TIM, and TIM₁₄₉. The results indicate that in the absence of TIM, the chip would reach temperatures exceeding 200 °C. However, with the inclusion of Commercial TIM and TIM₁₄₉, a significant temperature reduction is observed. Notably, TIM₁₄₉ maintains a temperature of 68.2 °C after stabilization, marking a substantial 152 °C decrease compared to the scenario without TIM. Fig. 5e presents the stable temperature profiles under different power levels, along with their respective linear fittings. Considering the typical operating temperature range for civilian chips (0–70 °C), it can be inferred that the TIM developed in this study functions effectively within a power density of 28 W/(cm²), whereas operation without TIM is limited to 3 W/(cm²). This enhancement significantly improves the power density and reliability of the chip. Furthermore, Fig. 5g investigates the long-term performance of the TIM under variable temperature cycles (60 s of operation at 30 W/(cm²), followed by 60 s of power-off, repeated for 1200 cycles). There is no outflow of powder and

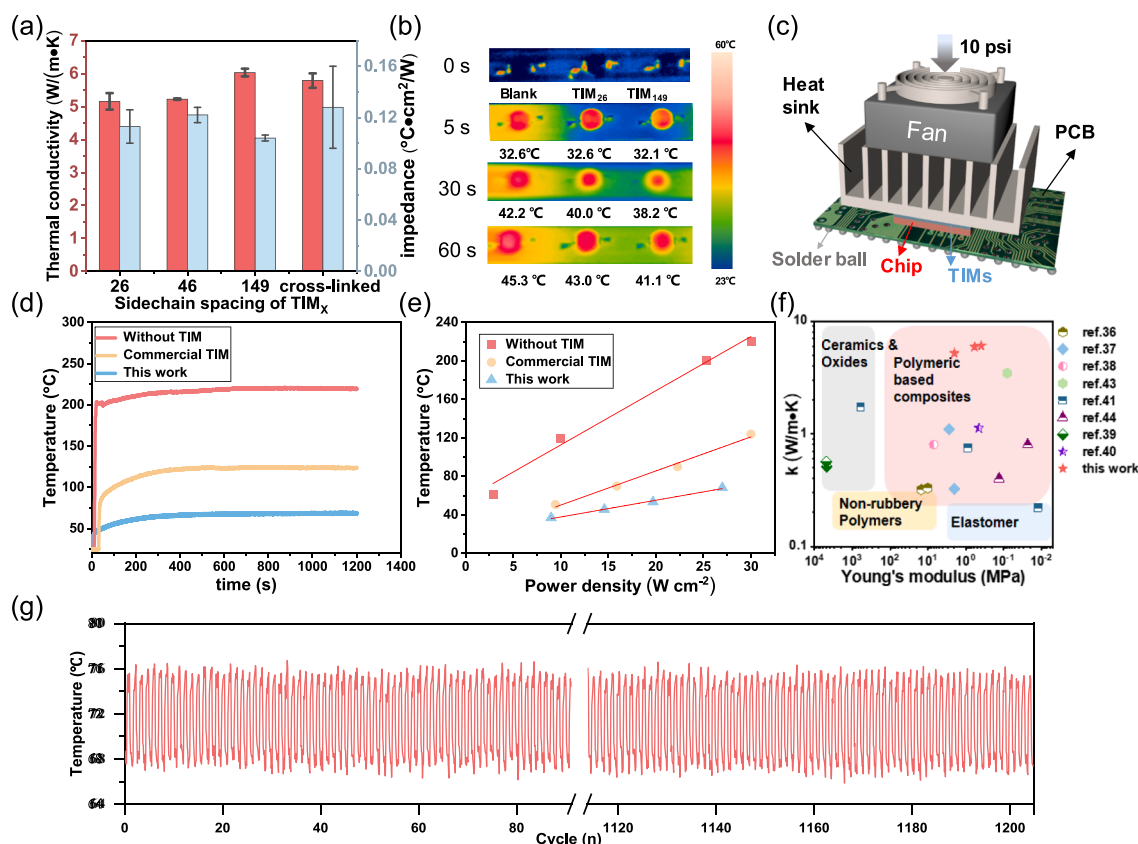


Fig. 5. Thermal conductive properties and application of thermal interface materials. (a) Thermal conductivity and impedance of the TIM_x (X is sidechain spacing, X = 26, 46, ...) with different dangling chain spacing. (b) LED lamp heat dissipation test. (c) Thermal management application of the TIMs in heat dissipation of CPU microprocessors (d) T3Ster coupled with shaking table tests on selected samples to continuously detect the real-time surface maximum temperature of chip. The heating time at an applied power of 30 W cm⁻². (e) T3Ster coupled with different power density. (f) Graphic comparison of the Young's modulus and thermal conductivity of this work with those of reported elastomers. (g) Thermal cycling stability of TIM₁₄₉ (input power 30 W).

matrix in temperature after more than 1200 cycles, which proves that the material has a certain reliability (Fig. S10). The results reveal low operating temperatures and robust stability at this power level. Any slight temperature drop in the later cycles is attributed to the thinness of the material and its minimal thermal resistance under prolonged pressure. In Fig. 5f, a comparative analysis is conducted with previously reported materials, further emphasizing the distinctive attributes of our composites [35–40,41,42]. In general, our material exhibits outstanding characteristics, including a low modulus and high thermal conductivity. This not only facilitates enhanced conformation to the chip and heat sink interface, thereby reducing contact thermal resistance, but also serves to further augment the heat dissipation capacity of the system.

3. Conclusion

In conclusion, this work has fabricated a thermal interface material with low modulus and high thermal conductivity by manipulating multi-branched PDMS and ultra-high Al fillers. This work found that side chains can not only untangle but also intensify the entanglement of composite materials. A low modulus and high thermal conductivity thermal interface material was prepared. As the side-chain spacing decreases, entanglements increase between dangling chains, between dangling chains and cross-link chains, leading to an increase in the modulus of PDMS and enhanced energy dissipation. Upon the addition of aluminum powders, the interaction between chains and fillers generated. More dangling chains result in more entanglement and greater energy dissipation. Combining these competition of disentanglement and entanglement of dangling chain, we prepared a thermal interface material TIM₁₄₉ with high thermal conductivity (6.02 W/(m K)), low thermal resistance (0.104 (cm² °C)/W at 40 psi), and low modulus (0.39 MPa) at a high Al content of 92 wt%. Furthermore, this thermal interface material demonstrates excellent performance in heat dissipation applications, providing a new type of material for thermal management applications.

CRediT authorship contribution statement

Zhian Zhang: Data curation, Formal analysis, Writing – original draft. **Zhibin Wen:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Funding acquisition. **Jiashuo Sheng:** Investigation, Data curation. **Chen Zeng:** Investigation, Data curation. **Jianfeng Fan:** Methodology, Formal analysis. **Linlin Ren:** Visualization, Supervision, Project administration, Funding acquisition. **Xiaoliang Zeng:** Writing – review & editing, Visualization, Resources, Funding acquisition. **Rong Sun:** Resources, Project administration, Funding acquisition. **Peng Chen:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhibin Wen reports financial support and equipment, drugs, or supplies were provided by Chinese Academy of Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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